Shirisha Gajendra

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Education

Stony Brook University — Stony Brook, NY
Master of Science in Data Science | GPA: 3.58/4.0 — *Aug 2025 – May 2027*
- **Coursework:** Statistical Learning, Big Data Analysis, Data Analysis, Data Management, Statistical Computing

Sai Vidya Institute of Technology — Bengaluru, India
B.E. Computer Science & Engineering (Data Science) | GPA: 9.17/10 — *Dec 2021 – May 2025*

Experience

Stony Brook University – Division of Nephrology and Hypertension — Stony Brook, NY
Research Assistant – Statistical Analysis — *Jan 2026 – Present*
- Conduct end-to-end statistical analysis for clinical comparative effectiveness research on electronic health record (EHR) cohorts of tens of thousands of patients, from cohort construction to final results.
- Apply causal inference (propensity score matching) and survival analysis (Cox proportional hazards, competing risks) in R to estimate treatment outcomes from real-world observational data.
- Lead an independent cognition module for a national semaglutide target-trial emulation in Python: profiled the cohort, mapped overlap across 6 cognitive outcomes, and built date-quality checks with 29 passing automated tests.
- Deliver manuscript-ready tables, figures, and summaries for 3 faculty-led, peer-review-track studies.

Rail Wheel Factory — Bengaluru, India
Data Analytics Intern — *May 2024 – Aug 2024*
- Built interactive Power BI dashboards tracking KPIs across 3 departments, giving managers and stakeholders a single view of operational performance.
- Automated SQL ETL pipelines that reduced manual reporting workload by 40%.

QSpiders Global — Bengaluru, India
Software Engineer Intern — *Sep 2023 – Nov 2023*
- Developed 2 Java applications using OOP and REST APIs; optimized SQL queries in an Agile team using Git.

Projects

A/B Test Analysis of a 14M-User Ad Experiment | *Python, SQL (DuckDB), statsmodels, Bayesian* — GitHub
- Analyzed a randomized ad experiment (Criteo, 14M users) in SQL and Python, validated randomization with an SRM test, and estimated a 59% conversion lift (95% CI 54–65%), confirmed by a Bayesian model.
- Ran power and minimum detectable effect analysis, cut variance 25% with cross-fitted CUPED, and diagnosed a covariate imbalance that shifted adjusted estimates, reporting a robust effect range.
- Separated intent-to-treat from the effect on exposed users (instrumental variable), showed a naive comparison overstated impact by a third, and wrote a break-even decision memo.

ReviewGuard – Explainable Fake-Review Detection | *PySpark, Spark MLlib, GraphFrames, SHAP* — GitHub
- Built a PySpark pipeline on YelpChi (46K reviews) and added reviewer graph features (degree, PageRank) with GraphFrames, raising AUC from 0.77 to 0.85 and doubling fraud recall over a logistic regression baseline.
- Flagged 548 suspected review campaigns via connected components and explained predictions with SHAP.

ResumeFit – Job-Market Analytics & LLM Agent | *Python, pandas, SQLite, OpenAI API* — GitHub
- Built a job-market analytics pipeline that scans 78 company job boards and alert emails every 5 minutes, stores postings in SQLite, filters by visa sponsorship and seniority, and surfaces the most in-demand skill gaps.

Skills

Languages & CS: Python, R, SQL, Java, C, C++, Data Structures & Algorithms
Statistics & ML: A/B Testing, Experimentation, Power Analysis, CUPED, Bayesian Statistics, Probability, Causal Inference, Target Trial Emulation, Real-World Evidence, Cohort Analysis, Data Quality, Econometrics, Forecasting, Product Metrics, Survival Analysis, Statistical Modeling, Predictive Modeling, Hypothesis Testing, Regression, Classification, Random Forest, Fraud Detection, Graph Analytics, Explainable AI (SHAP), Machine Learning, Feature Engineering, Data Visualization, Deep Learning (CNNs), NLP, LLMs
Libraries: pandas, NumPy, scikit-learn, PySpark, Spark MLlib, GraphFrames, tidyverse, Matplotlib, TensorFlow/Keras
Tools: Distributed Computing (Spark, MPI), DuckDB, statsmodels, Power BI, Tableau, MySQL, SQLite, Git, pytest, Jupyter, Flask, Streamlit, OpenAI API
Leadership: Vice President, Robotics Club (2023–2025), organized AI/ML hackathons and mentored 50+ students